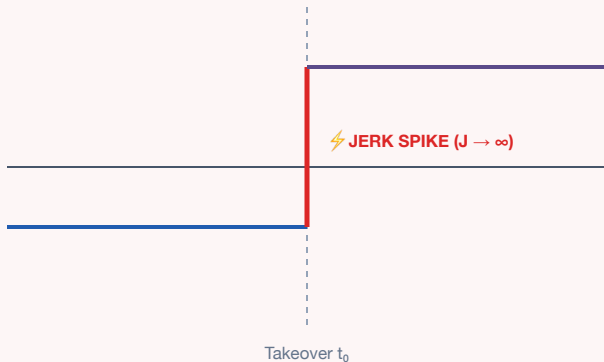


BUMPLESS TRANSFER: STEP DISCONTINUITY VS C² QUINTIC BLENDING

Eliminating Mechanical Jerk Shock During Human Takeover via $S(\alpha) = 10\alpha^3 - 15\alpha^4 + 6\alpha^5$

× STEP DISCONTINUITY (Naïve Hard Switch)

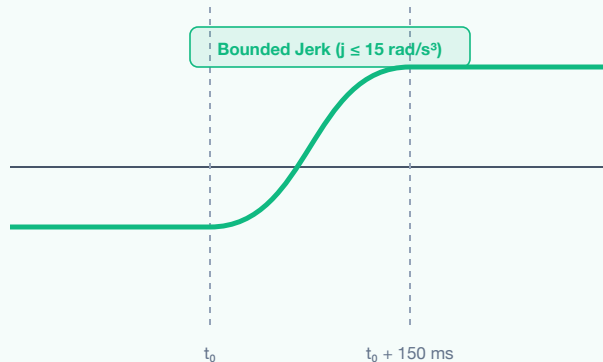
Instantaneous authority handover $u_t = u_{\text{human}}$



Gearbox tooth impact · Joint shock oscillation · Mechanical fatigue

✓ C² QUINTIC SMOOTHSTEP BLENDING

$u_t = (1 - S(\alpha)) \cdot p_t + S(\alpha) \cdot u_{\text{human}}$ ($\Delta t = 150$ ms)



Continuous acceleration · Zero contact bounce · Seamless operator feel